

University of Central Punjab

Faculty of Information Technology and Computer Science

Playfair Cipher

A digraph substitution cipher using a 5x5 matrix constructed from a keyword.

Steps:

1. Create 5x5 matrix from keyword (merge I/J).
2. Split plaintext into digraphs.
3. Apply following rules:
 - Same row → replace with right letter.
 - Same column → replace with below letter.
 - Otherwise → replace with letters at rectangle corners.

Task1:

- Implement encryption and decryption functions.
- Test with a suitable 5x5 key matrix on plaintext **"HELLO"**.
- Values must not be hardcoded.

Task2:

- Implement encryption and decryption functions.
- Test with keyword **"KEYWORD"** on plaintext **"INFORMATION"**.
- Values must not be hardcoded.
- Display the generated 5x5 matrix along with the encrypted and decrypted text.